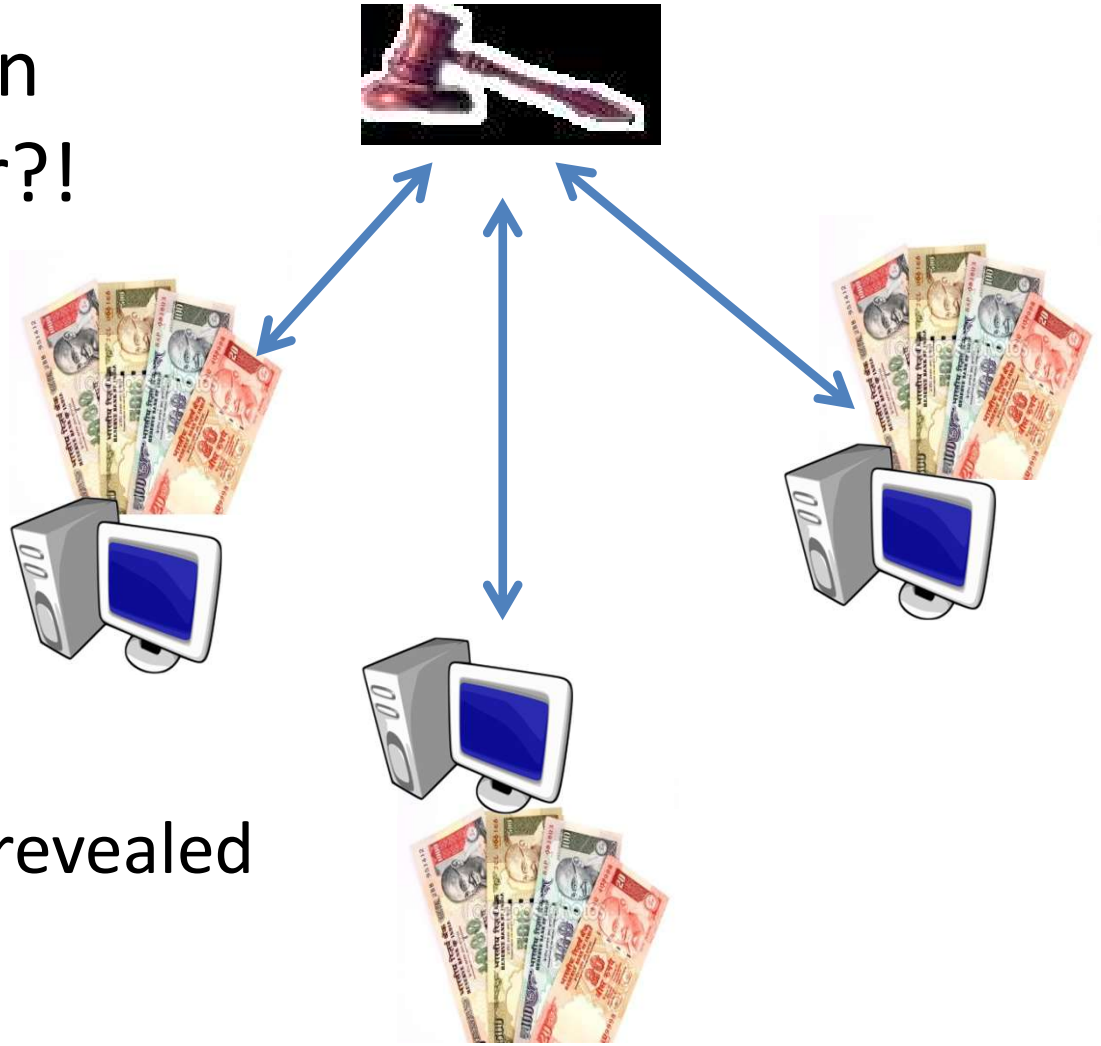
Secure Multiparty Computation in Cloud Security

Electronic Auction

- Can we have an auction without an auctioneer?!
  - Declared winning bid should be correct
  - Only the winner and winning bid should be revealed



Using data without sharing?

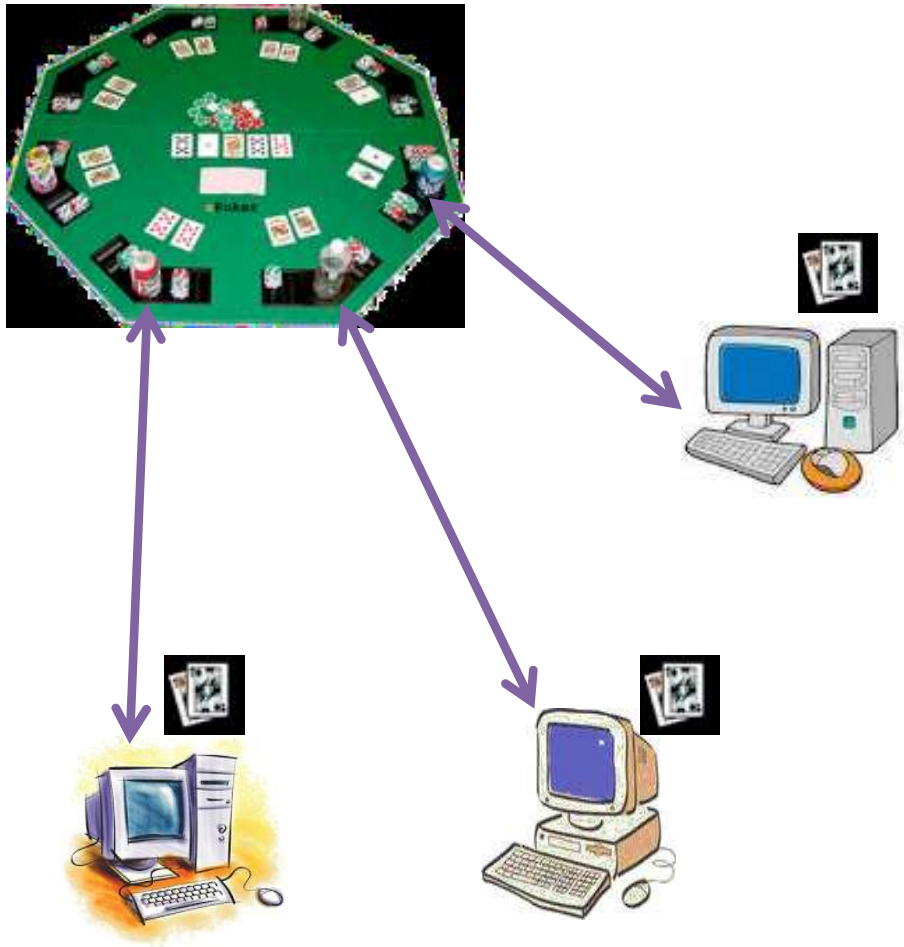
- Hospitals which can't share their patient records with anyone
- But want to data-mine on combined data



Poker with No dealer

No universally trusted Dealer

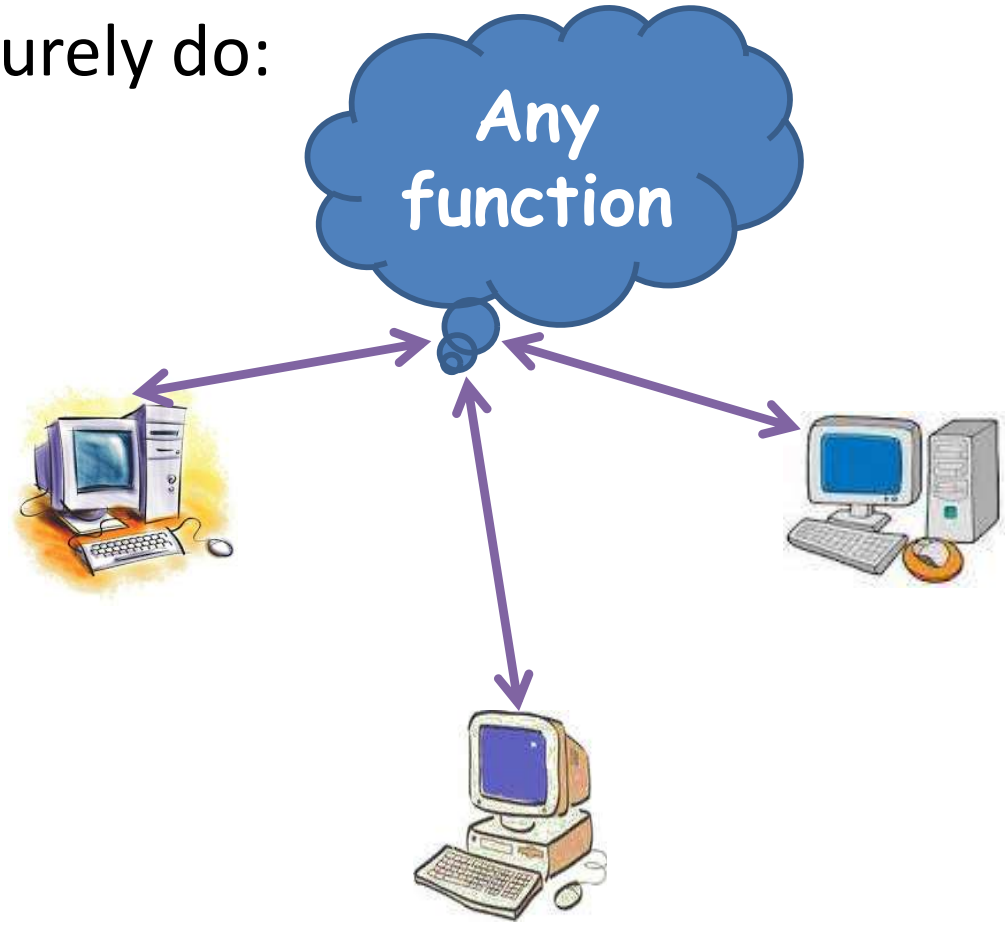
- Need to ensure:
  - Cards are shuffled and dealt correctly
  - Complete secrecy
  - No cheating by players, even if the collude



Goal

Without any trusted party, securely do:

- Distributed data mining
- E-commerce
- Network games
- E-voting
- Secure function evaluation
- ……



SMPC

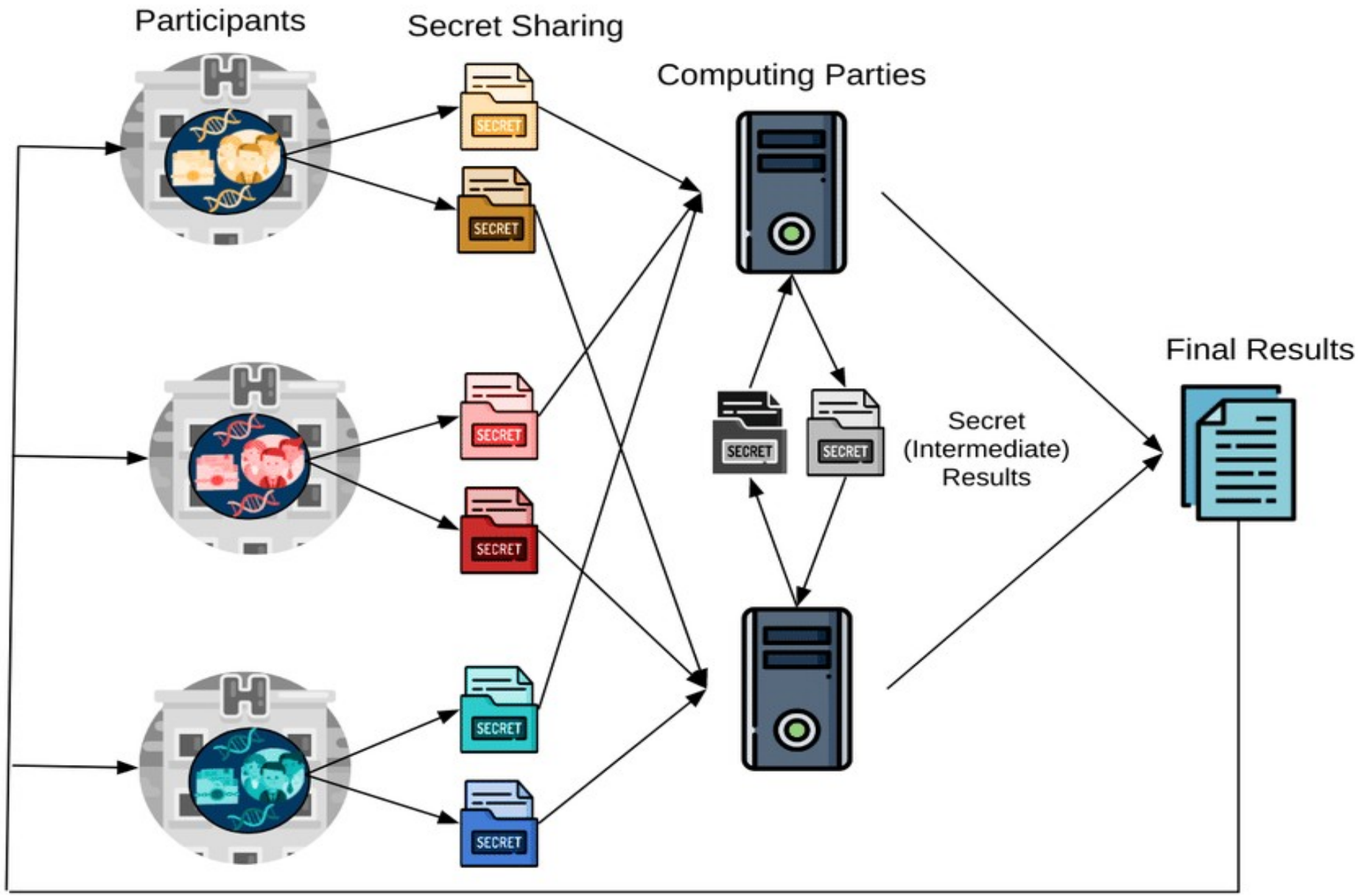
- A set of parties with “private” inputs wish to compute some “joint function” of their inputs.
- Secure protocol must reveal no more information than the output of the function itself.

SMPC

- The protocol execution may come under “attack” by an external entity or even by a subset of the participating parties.
- The aim of this attack may be to learn private information or cause the result of the computation to be incorrect.
- The two important requirements on any secure computation protocol are privacy and correctness.

- Privacy requirement states that nothing should be learned beyond what is absolutely necessary (parties should learn their o/p and nothing else).
- Correctness requirement states that each should receive its correct o/p.
- For example, In electronic auction protocol, privacy :only the winning bid is revealed.
- correctness: highest bidder is indeed the party to win.

Secure Multi-Party Computation

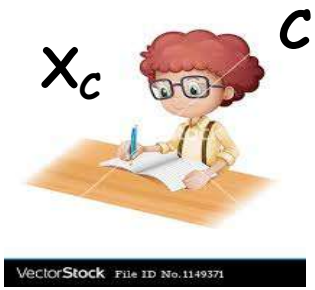


Summation protocol

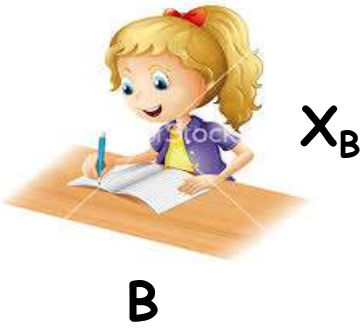
$X'_A = X_A$

$X'_B = X_B$

$X'_C = X_C$





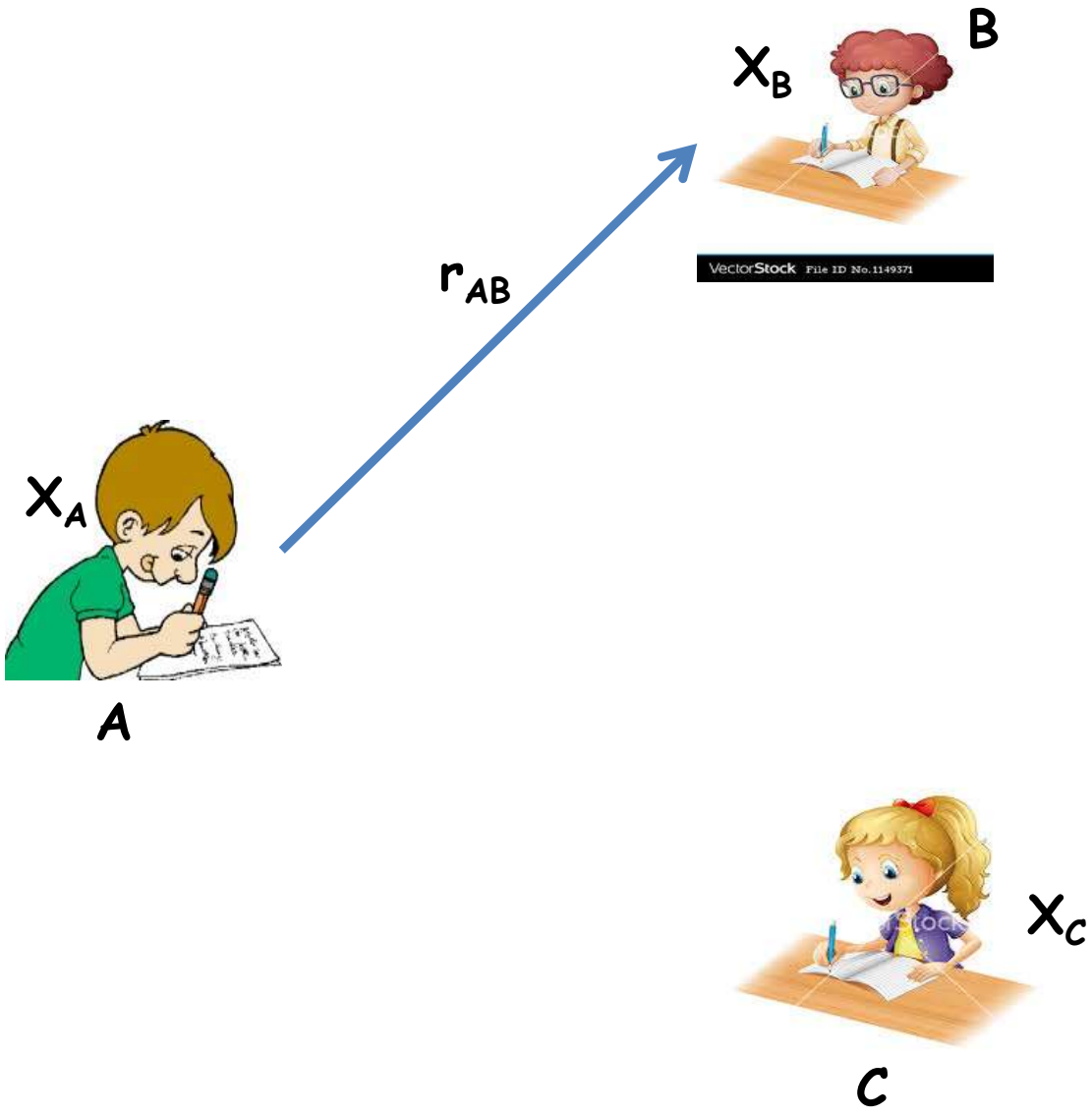


Summation protocol

$X'_A = X_A - r_{AB}$

$X'_B = X_B + r_{AB}$

$X'_C = X_C$

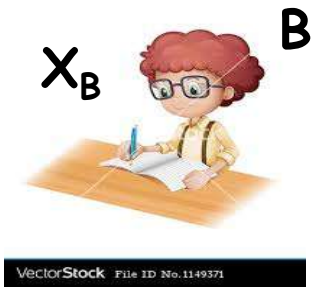


Summation protocol

$X'_A = X_A - r_{AB} - r_{AC}$

$X'_B = X_B + r_{AB}$

$X'_C = X_C + r_{AC}$





A

r_{AC}



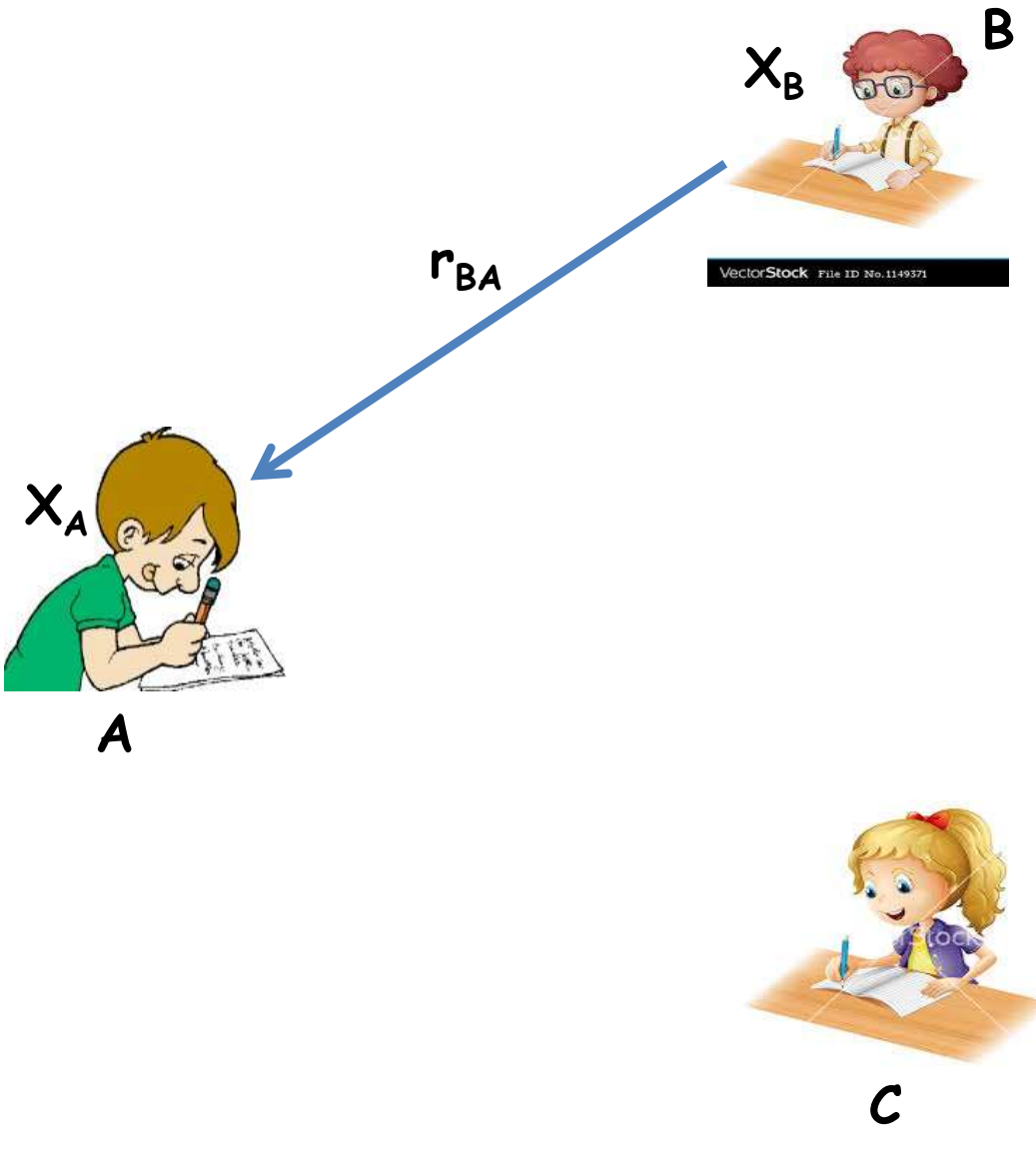
C

Summation protocol

$X'_A = X_A - r_{AB} - r_{AC} + r_{BA}$

$X'_B = X_B + r_{AB} - r_{BA}$

$X'_C = X_C + r_{AC}$



Summation protocol

$X'_A = X_A - r_{AB} - r_{AC} + r_{BA}$

$X'_B = X_B + r_{AB} - r_{BA} - r_{BC}$

$X'_C = X_C + r_{AC} + r_{BC}$



A



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r_{BC}



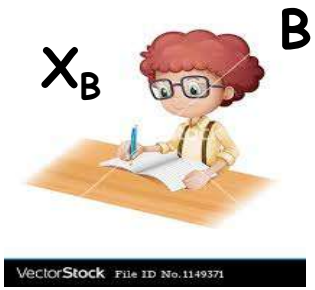
C

Summation protocol

$X'_A = X_A - r_{AB} - r_{AC} + r_{BA} + r_{CA}$

$X'_B = X_B + r_{AB} - r_{BA} - r_{BC}$

$X'_C = X_C + r_{AC} + r_{BC} - r_{CA}$





A

r_{CA}



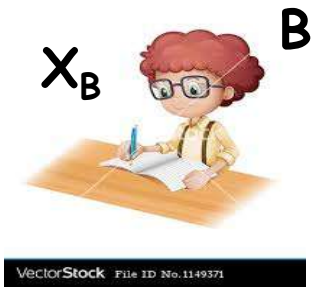
C

Summation protocol

$$X'_A = X_A - r_{AB} - r_{AC} + r_{BA} + r_{CA}$$

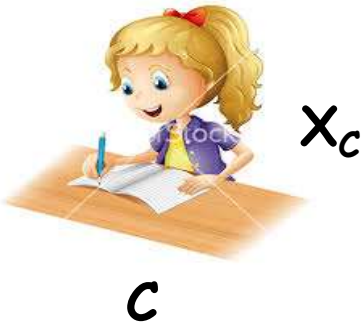
$$X'_B = X_B + r_{AB} - r_{BA} - r_{BC} + r_{CB}$$

$$X'_C = X_C + r_{AC} + r_{BC} - r_{CA} - r_{CB}$$





r_{CB}

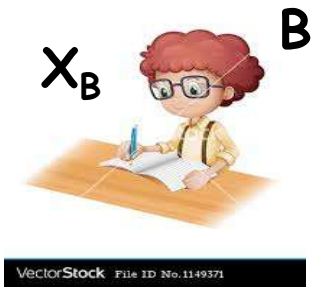


Summation protocol

$$X'_A = X_A - r_{AB} - r_{AC} + r_{BA} + r_{CA}$$

$$X'_B = X_B + r_{AB} - r_{BA} - r_{BC} + r_{CB}$$

$$X'_C = X_C + r_{AC} + r_{BC} - r_{CA} - r_{CB}$$





A

$$F(X'_A, X'_B, X'_C)$$

$$= X'_A + X'_B + X'_C$$



C

This problem takes many different forms depending on the underlying network, on the function to be computed, and on the amount of distrust the parties have in each other and in the network.

Cloud Computing

- Consider a cloud service storing encrypted images.
- Law enforcement may require the cloud to search for images containing a particular face.
- Thus, the cloud needs a restricted secret key that decrypts images that contain the target face, but reveals nothing about other images.
- More generally, the secret key may only reveal a function of the plaintext image,
  - For example an image that is blurred everywhere except for the target face.
- Traditional public-key cryptography cannot help with such tasks.

SMPC

Challenges in Cloud Computing

- Data Protection: Securing your data both at rest and in transit
- User Authentication: Limiting access to data and monitoring who accesses the data
- Disaster and Data Breach Contingency Planning

Providing Secure storage and treatment of data in cloud environment securely is the main goal

Role of SMPC in Cloud Computing

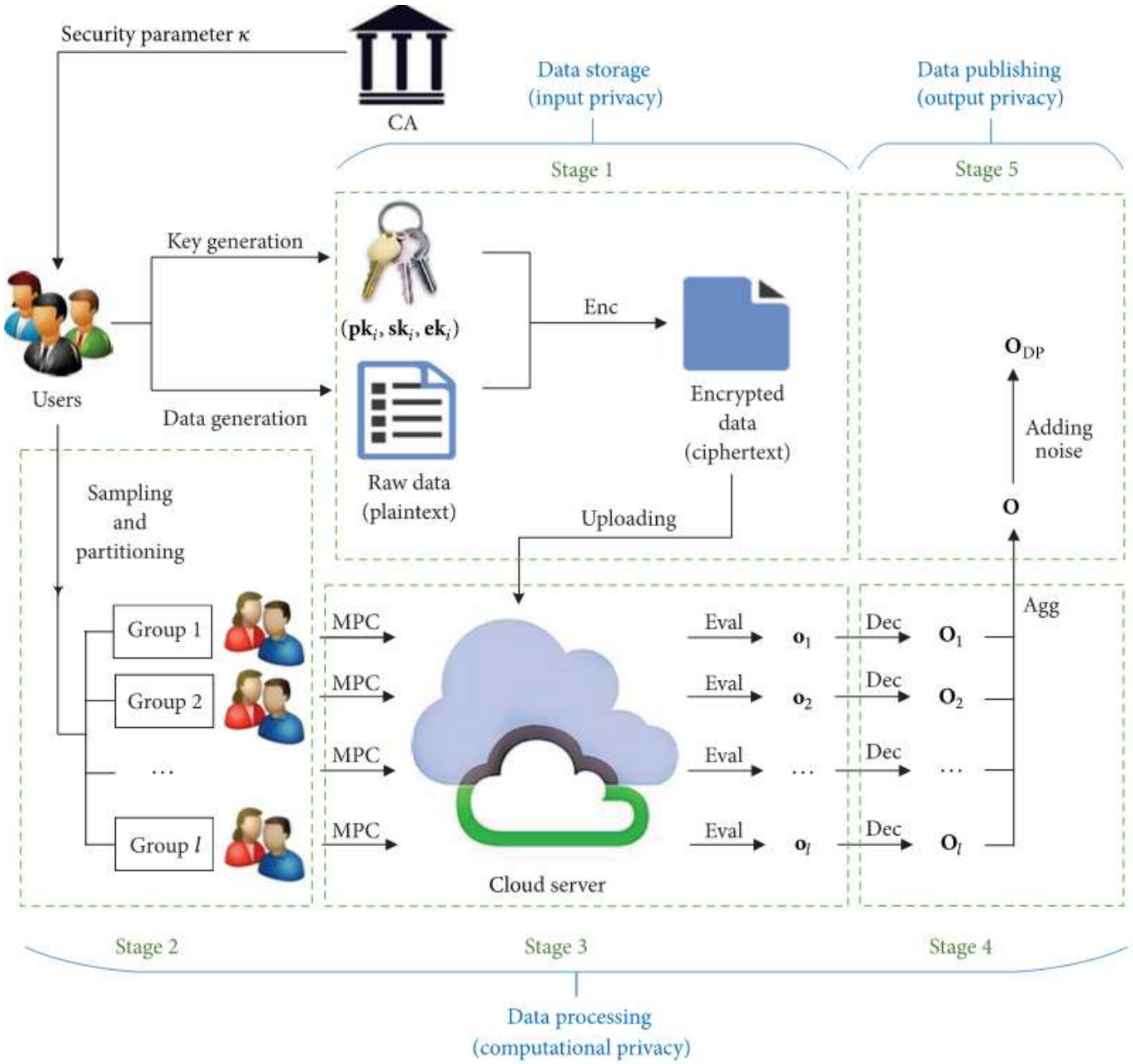
Example

- Consider a case where multiple cloud users jointly work on a project and utilize the services of the same cloud provider.
- All of them want to simultaneously evaluate their equations without revealing individual equations to one another.
- In such scenarios the SMPC solutions can be highly useful to provide privacy.

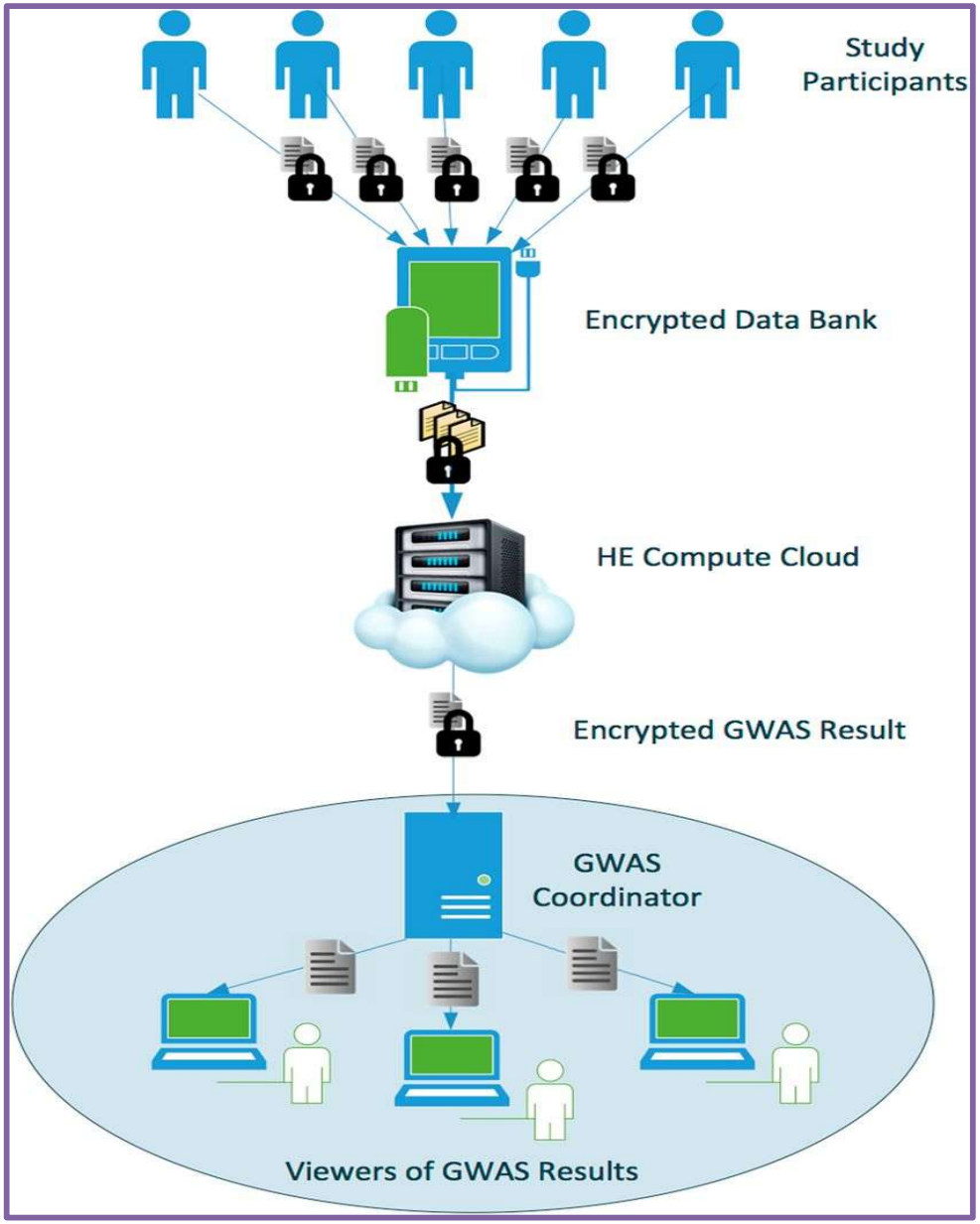
Solution

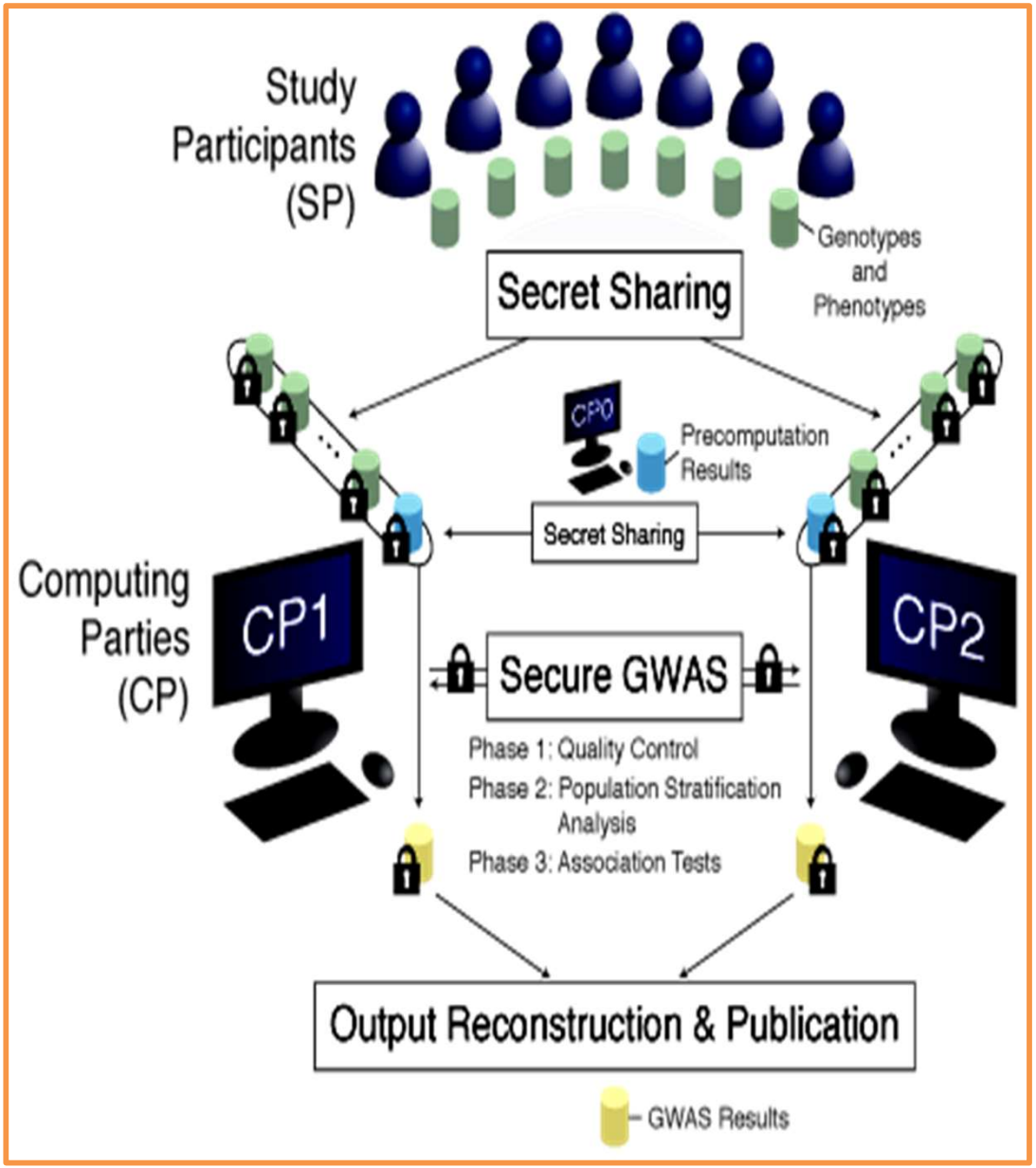
SMPC solution techniques can be embedded while designing architecture of cloud computing especially when multiple cloud providers or multiple cloud users jointly compute some function of their private data inputs.

SMPC in Cloud



SMPC Applications





SMPC Applications

Fraud Data Detection in Banks

- SMC enables the financial services organization to train their risk and fraud models on distributed data from participating partner banks while preserving the privacy and security standards of all partner banks.
- So that the predictiveness of their fraud detection algorithms can be increased.

Health Sector

- In its cardiovascular disease research programmes, researchers/ healthcare professionals have access to more features and samples, resulting in more reliable disease prognosis.
- Increased use of private data by a large healthcare provider aims to aid early diagnosis of cardiovascular ailments .
- Contributors are prohibited from disclosing data outside the organizational gates due to privacy rules such as confidentiality issues and Health Insurance Portability and Accountability Act (HIPAA).

Limitations and advantages

- There is a lot of communication overhead.
- random numbers must be generated- might slow down the execution time.
- Secret sharing involves connectivity and communication among all participants, which leads to higher communication costs as compared to plaintext computation.
- Not vulnerable to adversaries with high computational power
- No need of any trusted third party
- To protect the privacy of data, there is no need to disguise or remove any elements.

THANK YOU